No of Pages : 2 **Course Code : 23MX23**

Roll No:
(To be filled in by the candidate)

PSG COLLEGE OF TECHNOLOGY, COIMBATORE - 641 004

SEMESTER EXAMINATIONS, AUGUST / SEPTEMBER 2024

MCA Semester: 2

23MX23 OBJECT ORIENTED PROGRAMMING USING JAVA

Time : 3 Hours **Maximum Marks :100**

INSTRUCTIONS:
1. Answer **ALL** questions. Each question carries 25 Marks. 2. Subdivisions (a) carries 3 marks each, subdivision (b) carries 10 marks each and subdivision (c) carries 12 marks each.

1. a) Comment on Java Bytecode. How does it helps on providing platform independent?

   b) i) Write a short note on implicit memory deallocation and explicit memory deallocation with example code snippet.

   ii) Can we explicitly deallocate an object from memory in Java? How Java language treats object vs primitive variables while passing as argument?

   c) Explain in detail about architecture of Java virtual machine. What will happen if null value is assigned explicitly to the reference type in java program?

2. a) What is the significant use of the keyword 'this' in Java?

   b) i) In object-oriented programming, state what is class, instance, method, and message and give an example of each.

   ii) Can a class in Java be inherited from more than one class? Justify with relevant explanations about how java handles multiple classes.

   c) i) Explain the multiple inheritance and interface. How is extending a class different from implementing an interface? Give an example.

   (OR)

   ii) With relevant examples, differentiate method overloading and method overriding. Justify relevantly how method overriding supports the concept of Interface implementations especially in multiple inheritance.

3. a) What is multi-threading? Why Runnable Interface is used in Java?

   b) Elaborate various stages on the life cycle of a thread. Present any three scenarios that state how the programmer needs to choose sleep and/or wait method. State its reason.

c) i) Explain inter thread communication and synchronization in Java and state how it is accomplished in Java programming languages with suitable example?

(OR)

ii) Explain the difference between the Reader/Writer class hierarchy and the InputStream/OutputStream class hierarchy. A hotel salesperson enters sales in a text file. Each line contains the following, separated by semicolons: The name of the client, the service sold (such as Dinner, Conference, Lodging, and so on), the amount of the sale, and the date of that event. Write a program that reads such a file and displays the total amount for each service category. Display an error if the file does not exist or the format is incorrect

4. a) What is significance of using Arraylist?

b) State the significance of Java Collections and Packages with examples. With relevant examples, write a note on Sets and Maps collection.

c) Explain in detail event handling mechanism in Java. List any four event classes and its listener interfaces.

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FD/JU